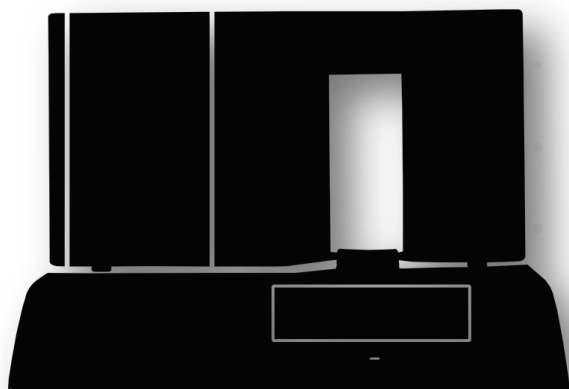


Luminex® 200 RoHS Compliant

L-Block Configuration



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General Observations

Cuvette's Breaking During Shipping

Observed systems with a cracked cuvette within the bird cage. The crack resulted in sheath flooding of the optical bench and analog board. To minimize liquid damage, please pop the covers and inspect cuvette before turning the power on.

MagCAL1 & MagCON1

On initial run, MagCAL1 and MagCON1 must be run in LXMaint 3.2.111 **BEFORE** running CALs and CONs when performing calibration. This is a known software bug that will cause calibration failures otherwise.

LxMaint 3.2.111

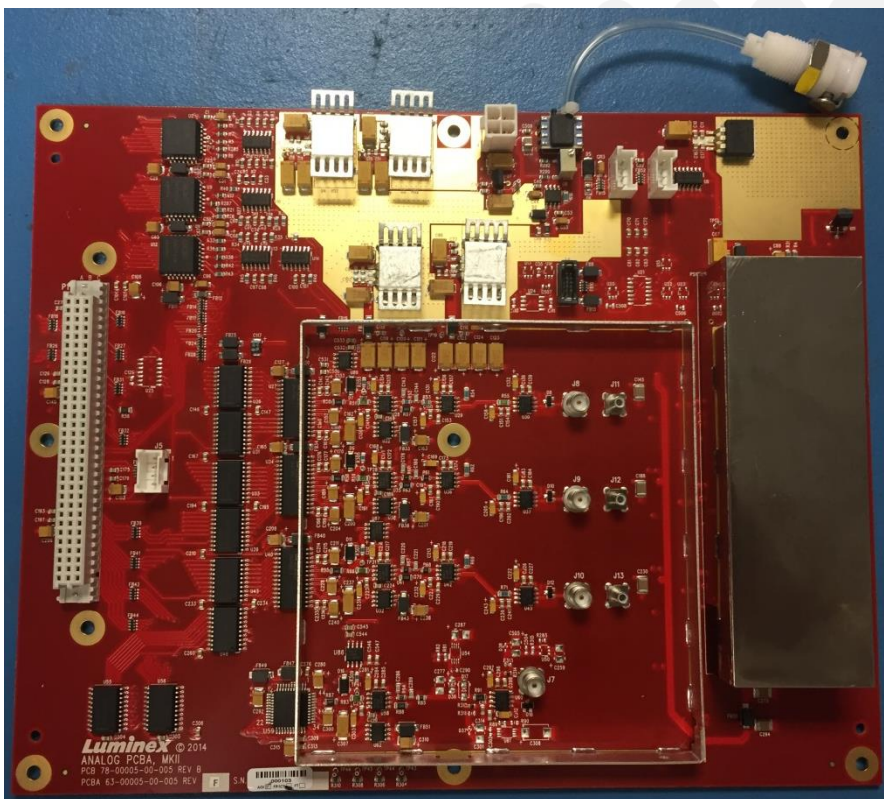
Maintenance software tends to get delayed during fine tuning of alignment beads. When sample is nearing end of run, CVs and dot plot looks stalled. Once new sample is run, software resumes as normal.

Board/Kit Information

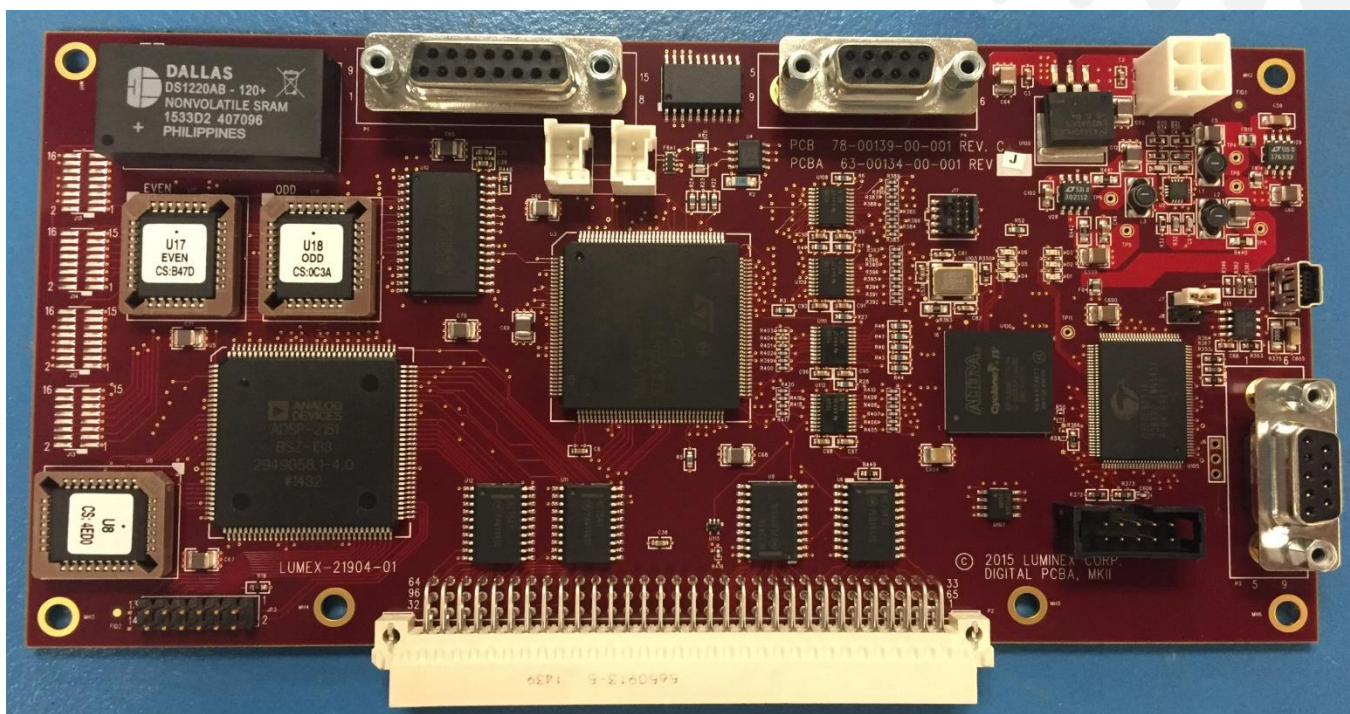
	Board PN	Retro Fit Kit PN
Analog Board	FS-1133-01	FS-1134-01
Digital Board	FS-1135-01	FS-1136-01
Laser Driver Board	FS-1137-01	FS-1138-01
Sensor Distribution Board	FS-1144-01	FS-1145-01

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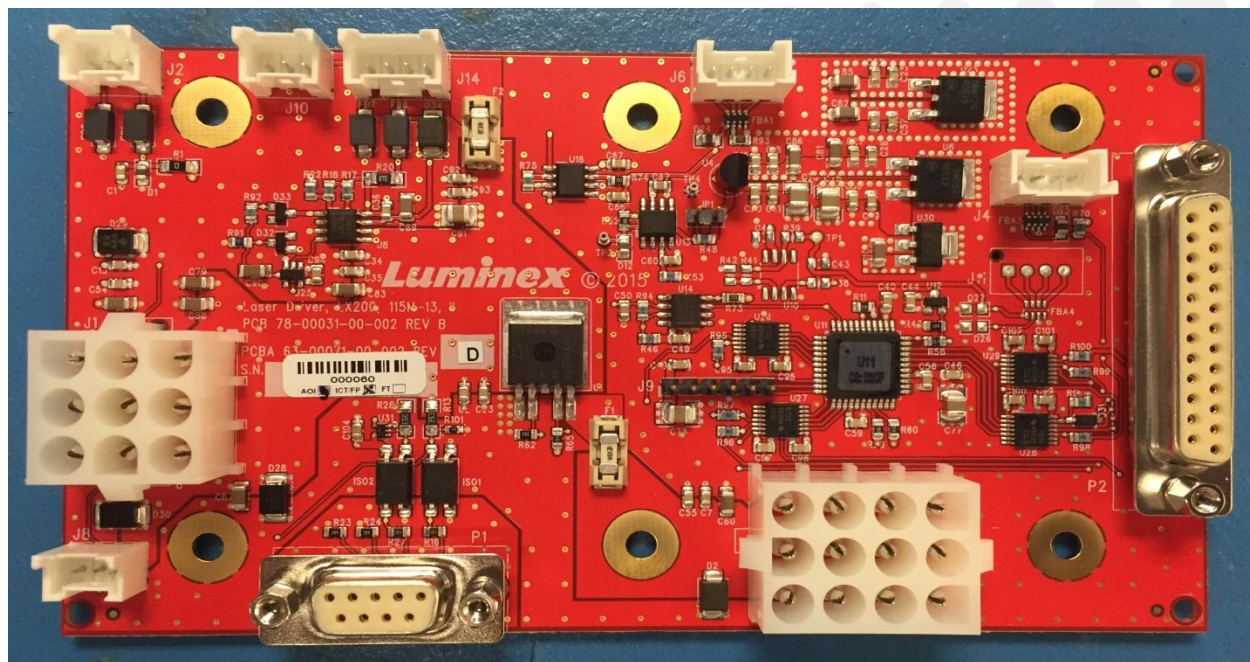
Analog Board



Digital Board

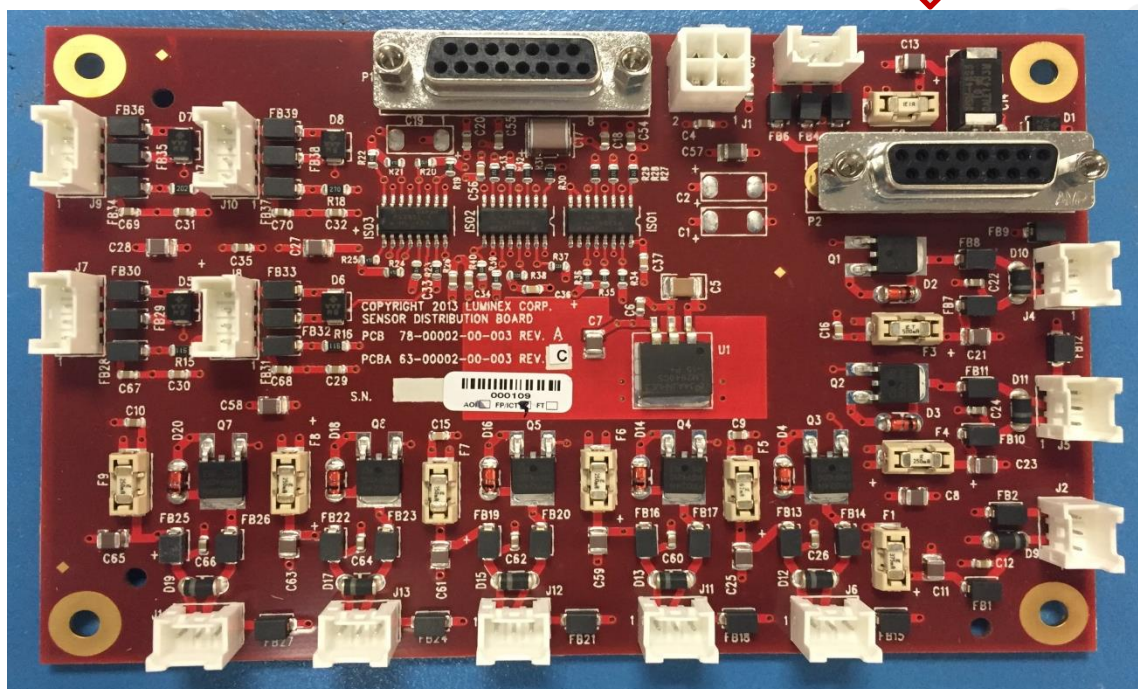


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**Laser Driver
Board**

**Sensor Distribution
Board**



Optical Alignment Procedures

Issues with CVs only: (alignment only)

1. Run CAL1/Alignment beads and perform fine tuning.

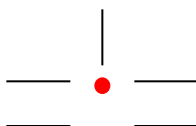
Red Laser Replacement:

NOTE: *It is worthwhile to attempt side to side and fine tuning when replacing the red laser prior to the full alignment effort. Laser boresight may vary between lasers, hence the procedure listed below.*

1. Remove red laser focus assembly and classification dichroic mirror.
2. Remove defective laser and plate as a single assembly, two screws on top and two screws below the platform.
3. Gap the red laser focus assembly, 10mm lens, reporter and classification dichroic mirrors, and both 4mm lenses. Use either a gapping tool or a 1/8th Allen wrench.
4. Remove L-block from the platform and remount on the rear mounting screw.
5. Install the new red laser and plate assembly, snugging, but not tightening the laser mounting plate (4 total screws).
6. With the analyzer on, turn off the green laser, leaving the red laser on, and place a card in front of green laser. (Leaning the card may make it easier due to the low position of the red laser dot with the 10mm and red focus assembly removed.)
7. Move the red laser and plate assembly side to side to center on cuvette. (It will help to tilt the card to lean against the green laser since spot will be low.) Tighten the mounting screws while holding the plate firmly in place.
8. Install the red laser focus assembly, snug, but not tight (2 screws).
9. Again, move the red laser and plate assembly side to side to ensure laser is still focused on cuvette. Tighten the mounting screws while holding the plate firmly in place.
10. Loosen the green laser just enough to give small movement.
11. Turn off the red laser and turn on the green laser, placing a card in front of red laser focus assembly.
12. Move the green laser side to side to center on cuvette. Tighten the mounting screws while holding the laser firmly in place.

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13. Install the 10mm lens tight against the screws. Ensure both lasers are turned on.
14. Adjust the vertical screw on the 10mm lens to place the red dot just on top of the green dot on a card against the green laser.
15. Install the reporter, classification and I-block test jigs. (Ensure the L-block test jig is held tightly to the side of the platform when tightening the screw.)
16. Set up a long sample with the following settings (LxMaint 3.2.111):
 - Acquisition Tab
 - Acquisition Set-up
 - Advanced Settings Tab
 - Uncheck DD
 - Auto-stop Setup: None
 - Time (sec): Max 200
 - Sample Size (μL): 200
17. Turn green laser off and place a card in front of PMT. Set RP1 gain to zero.
18. Run Red Laser Alignment Suspension as a sample. ***Be sure to disable temperature compensation and uncheck high voltage.***
19. Align red dot using the classification 4mm lens to target upper crosshairs on classification test jig. Perform focal turns on the red laser focus assembly as needed to achieve clear edges on dot.

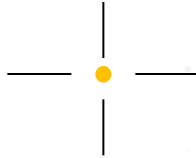


20. Run 2 alcohol washes and block the PMT.
21. Run Laser Alignment Solution as a sample. ***Be sure to disable temperature compensation and uncheck high voltage.***
22. Align yellow dot using the 10mm lens to target lower crosshairs on the ***classification*** test jig. Perform focal turns on this lens as needed to achieve clear edges on dot.



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23. Align the yellow dot using the reporter 4mm lens to the crosshairs on the **reporter** test jig. Perform focal turns on this lens as needed to achieve clear edges on dot.



24. Remove classification and reporter test jigs.
25. Install classification and reporter dichroic mirrors tight against the screw shafts.
26. Place a white card behind the L-block test jig.
27. Run Red Laser Alignment Suspension as a sample. **Be sure to disable temperature compensation and uncheck high voltage.**
28. Adjust the classification dichroic mirror to center the red dot through the two reference holes on the L-block test jig.
29. Install the L-block with test jig present. Tighten the L-Block and remove the test jig.
30. Run Laser Alignment Solution as a sample. **Be sure to disable temperature compensation and uncheck high voltage.** With the RP1 Gain set to zero and high voltage turned off, it is safe to remove the card covering the PMT to aid in centering.
31. Adjust the reporter dichroic mirror to center the yellow dot over the PMT opening. Use the vertical screw on the 4mm lens to adjust the shape of yellow dot. Replace the card in front of the PMT when finished.
32. Run 2 alcohol washes. Ensure the RP1 gain is set to zero .
33. Set up O-scope with the following settings on the analog board:
- Channel 1: DD
 - Channel 2: RP1
 - Trigger on channel 2, rising slope and set trigger just above signal trace
 - 10 μ S/division for both channels
 - 500mV/division for both channels
34. Set RP1 gain to 500. Remove card from PMT.
35. Run Alignment Microspheres as a sample. Optimize the classification side by using the dot plot and oscilloscope to maximize oscilloscope signal amplitude and tighten dot plot grouping as much as possible.

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36. Accomplish this with small adjustments to classification dichroic mirror and red laser focus assembly screws while monitoring both the oscilloscope and dot plot. The dot plot signal should be as high and as far right as possible. Configure dot plot axis to:
 - x-axis: CL1
 - y-axis: CL2
37. Repeat the previous step on the reporter side touching the dichroic mirror and 4mm lens screws. Configure dot plot axis to:
 - x-axis: DD
 - y-axis: Reporter
38. On the oscilloscope, select “Measure” and set for CH2 Pulse Width. The Reporter signal should measure 4.55-4.85uS if properly focused. Focal turns on the 10mm lens will help with this adjustment.
39. Once satisfied, run MagCAL1 and MagCON1 **first**, then CAL1, CAL2, CON1, CON2 to test alignment. All tests should pass.
 - If CVs fail on the classification side, perform focal turns on the red laser focus assembly and retest.
 - If CVs fail on the reporter side, perform focal turns on the 10mm lens and retest.
 - If CONs fail on either side, perform a stringent clean multiple times and retest all MagCAL/MagCON/CAL/CON beads again.

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Green Laser Replacement only:

1. Remove the defective green laser and 10mm lens.
2. Replace the green laser by tightening the two screws holding the laser in place.
3. With the laser on, place a card at the red laser focus assembly and turn off the red laser.
4. Move the green laser side to side until centered on the cuvette. Laying the card down against the red laser focus assembly can aid in centering. Be sure to minimize the comet tails to either side of the dot while moving the green laser side to side.
5. While holding the green laser firmly in place, tighten the two mounting screws.
6. Install the 10mm lens and center the green laser on the cuvette using the x-axis adjustment.
7. Run Laser Alignment Solution as a sample. ***Be sure to disable temperature compensation and uncheck high voltage.*** With the RP1 gain set to zero and high voltage turned off, it is safe to remove the card covering the PMT to aid in centering.
8. Adjust the reporter dichroic mirror to center the yellow dot over the PMT opening. Use the vertical screw on the 4mm lens to adjust the shape of yellow dot. Replace the card in front of the PMT when finished.
9. Run an alcohol wash and two water washes.
10. Set up O-scope with the following settings on the analog board:
 - Channel 1: DD
 - Channel 2: RP1
 - Trigger on channel 2, rising slope and set trigger just above signal trace
 - 10 μ S/division for both channels
 - 500mV/division for both channels
 - Set RP1 gain to 500. Remove card from PMT.
11. Run Alignment Microspheres as a sample. Optimize the reporter side by using the dot plot and oscilloscope to maximize oscilloscope signal amplitude and tighten dot plot grouping as much as possible. Accomplish this with small adjustments to reporter dichroic mirror and 10mm lens screws while monitoring both the oscilloscope and dot plot. The dot plot signal should be as high and as far right as possible.

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12. Configure dot plot axis to:
 - x-axis: DD
 - y-axis: Reporter
13. On the oscilloscope, select “Measure” and set for CH2 Pulse Width. The Reporter signal should measure 4.55-4.85 uS if properly focused.
14. Once satisfied, run MagCAL1 and MagCON1 **first**, then CAL1, CAL2, CON1, CON2 to test alignment. All tests should pass.
 - If CVs fail on the reporter side, perform focal turns on the 10mm lens and retest.
 - If CONs fail, perform a stringent clean multiple times and retest all MagCAL/MagCON/CAL/CON beads again.